DinHot: A Holographic Optical Trap Studio Comparing Classical, State-of-the-Art and Beyond-State-of-the-Art Hologram Methods on a Device Model

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github.com/fsantibanezleal/CEFOP_DinHot

Abstract

A phase-only spatial light modulator in the pupil of a microscope objective can form hundreds to thousands of optical traps at once, but only if the phase pattern it displays is computed well: the pattern fixes how much light reaches the traps, how equal the traps are, where stray light goes, how the pattern survives a real device, and how fast it can be recomputed when traps move. DinHot is an open studio that runs 24 ladder methods (10 classical, 9 state of the art, 5 beyond it) on the same task, with the same seeds, metrics and device model, in Python as the reference and in a TypeScript port that runs live in a browser. We describe the optical and device model, the methods with the measured acceptance of each, 14 cases from a 3×3 array to a disordered 819-trap array and shaped potentials, the benchmark protocol, two learned models, and the 5 proposals past the published state of the art, three of which meet their criteria (a device-aware weighted Gerchberg–Saxton update, a flicker-bounded phase-continuous sequence and an efficiency–uniformity front), one of which meets a narrowed criterion (an aberration-robust hologram) and one of which is refuted by measurement (a learned warm start for phase-fixed weighted GS). A network trained on three-plane volumes beats its baseline at equal wall-clock on all 162 held-out volumes (median accuracy ratio 1.140), whereas its two-dimensional counterpart ties with Gerchberg–Saxton (ratio 0.9994). The model is a scalar Fourier model of a simulated device, not a calibrated instrument.

Keywords: holographic optical tweezers, spatial light modulator, weighted Gerchberg–Saxton, pixel crosstalk, neutral-atom arrays, hologram benchmarking, learned holography.

1 Introduction and problem

Holographic optical tweezers shape one laser beam into many traps with a phase-only spatial light modulator (SLM). The largest published neutral-atom tweezer array holds 6,100 atoms in about 12,000 tweezers, with a trap-depth spread of 11.4 percent across its sites [1]. The spread matters: sites with unequal depth differ in loading and in gate fidelity, and the light that forms them is hard to keep uniform as the array grows. The same computation serves optical micromanipulation of beads and cells, neuron stimulation and laser machining, with different limits on power, speed and the number of traps.

The hologram methods that solve this problem span half a century: the superposition of prisms and lenses, random superposition and random masks, the Gerchberg–Saxton (GS) iteration and its adaptive and weighted forms [2], direct search [5], conjugate-gradient minimization [6], phase-fixed weighted GS for large arrays [3], mixed-region amplitude freedom for shaped potentials [4], gradient descent through differentiable forward models [7], optimal-transport initialization [15, 16], and learned holography [14]. Camera feedback closes the loop on a real set-up [8]. A search of the literature found no controlled published comparison of weighted GS,

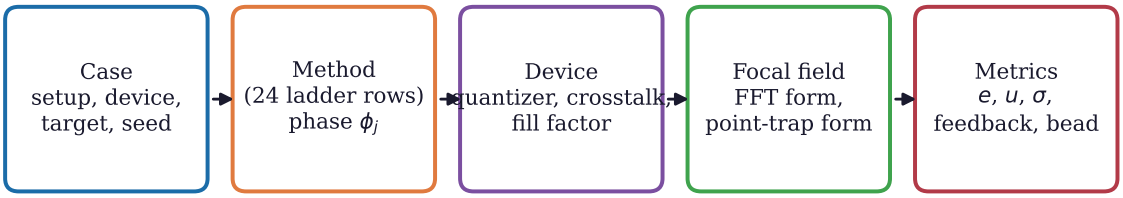


Figure 1: The chain every case runs: a case (optical setup, device, target and seed) is given to a method, which returns a phase; the device model turns the phase into the field the instrument forms; the metrics score the traps, and the feedback and particle simulations act on the scored field.

phase-fixed weighted GS, conjugate gradient, MRAF and a learned method on the same point-trap array, and no public tool that runs them side by side with a device model and the physics of what is trapped.

DinHot provides that comparison. It began at CEFOP, Universidad de Concepción, as DinHotSys, a C++ desktop tool for dynamic holographic optical tweezers dated 2010; it was rewritten in Python and FastAPI and is rebuilt here as a static browser application whose numbers are computed by the engine or baked offline and replayed. This report describes the problem, the optical model and the device, the three tiers of methods, the cases and the protocols, the results, the learned models, the proposals beyond the published state of the art, the live engine and the limits. Every number in the tables and the text is generated from the committed artifacts of the repository or measured by the scripts kept with this report; the browser timings are quoted from the product's record of measured resources.

2 The optical model and the device

2.1 Forward models

A phase-only SLM of $n_{\mathrm{pix}} \times n_{\mathrm{pix}}$ pixels with pitch p sits in the pupil plane of a lens of focal length f and wavelength λ. For a trap m at (x_m, y_m, z_m) with vortex charge l_m the phase delay at pixel j is

$$\Delta_j^m = \frac{\pi z_m}{\lambda f^2}(x_j^2 + y_j^2) + \frac{2\pi}{\lambda f}(x_j x_m + y_j y_m) + l_m \,\mathrm{atan2}(y_j, x_j), \tag{1}$$

and the trap field and intensity are

$$V_m = \frac{1}{N_{\mathrm{pix}}} \sum_j A_j e^{\,i(\phi_j - \Delta_j^m)}, \qquad I_m = |V_m|^2, \tag{2}$$

with A_j the input beam amplitude normalised so that $\sum_j A_j^2 = N_{\mathrm{pix}} = n_{\mathrm{pix}}^2$, following Di Leonardo et al. [2]. I_m is the fraction of the input power in the peak of trap m. The second form is the centred, zero-padded, orthonormal fast Fourier transform (FFT) of the SLM field; at a trap on an FFT bin with $z = 0$ and no vortex the two forms agree exactly, which the test suite checks to 10^{-9} rad. One bin of the padded transform spans $\lambda f/(\mathrm{pad}\, n_{\mathrm{pix}}\, p)$ and the unaliased half-window is $\lambda f/(2p)$; a trap outside it is refused with the bound (case C11). Depth is handled by the angular spectrum with the band limit of evanescent components removed; the model is scalar and paraxial, so a high numerical aperture is flagged as outside its validity (case C07).

2.2 Metrics

With I_m the trap intensities divided by their target weights, DinHot reports

$$e = \sum_m I_m, \qquad u = 1 - \frac{\max I - \min I}{\max I + \min I}, \qquad \sigma = 100\,\frac{\mathrm{std}(I)}{\mathrm{mean}(I)}\ \%, \tag{3}$$

the efficiency, the uniformity ($u = 1$ is perfect) and the relative standard deviation. Polin et al. [5] use the same ratio without the leading $1-$, a non-uniformity; the two are not mixed. A power budget separates

Table 1: The optical setups of the cases, read from the committed manifests and traces. The device row of C08 is the device the benchmark also uses for its device cells.

Setup	λ (nm)	f (mm)	NA	SLM pixels	pitch (μm)	pad	bin (μm)	half field (μm)
Atom (C01–C04, C09–C13)	850	6	0.5	256^2	20	2	0.498	127.5
Bead (C05–C07, water)	1064	2.4	1.2	256^2	10	2	0.499	127.7
C08	850	3	0.5	128^2	20	2	0.498	63.8
C14	850	2.56	0.5	128^2	20	1	0.850	54.4

The bead setup is in water of index 1.33; C14 is the volume of the released learned model. C08 device: 4-bit phase, crosstalk $\sigma = 1.24$, $q = 1.80$ on 3×3 sub-pixels, fill factor 0.85, flicker 0.05 rad.

the traps, the zero order, the ghost traps (everything else on the grid, including folded replicas) and the remainder.

2.3 The device

A real SLM quantizes the phase, couples neighbouring pixels, covers only part of each pixel and fluctuates. The device model applies, in order, wrapping, quantization to 2^b levels, a nonlinear response, Gaussian flicker of stated rms and wrapping, and blurs the upsampled phase with the crosstalk kernel

$$K = \mathcal{F}^{-1}\Big\{ \exp\Big[-\frac{|k_x|^q + |k_y|^q}{\sigma^q} \Big] \Big\}, \qquad \sigma = 1.24,\ q = 1.80, \tag{4}$$

the fitted parameters of Schroff et al. [9] on 3×3 sub-pixels (error below 2 percent in that work). A square pixel of fill factor fill multiplies the focal field by a product of cardinal sines, so the intensity falls to about 0.4 at the edge of the unaliased window; this envelope is derived in the engine and tested against a brute-force transform. The adjoint of the crosstalk operator is closed form (convolution with the reversed kernel and a block sum), which lets gradient methods differentiate through the device without automatic differentiation. Zernike aberrations follow the Noll indexing [10]. Table 1 lists the three optical setups of the cases.

3 The three tiers of methods

The design document fixes, before any code, a ladder of 24 rows: 10 classical methods (the established methods of the problem), 9 state-of-the-art methods (current solutions, replicated or implemented, never only cited) and 5 proposals beyond the state of the art. Each row has one measurable acceptance criterion and one test. Table 2 lists them; Table 3 gives what each test measured on DinHot's own grids, with the seeds stated.

3.1 Classical tier

Analytic baselines (M01 to M03). The superposition of prisms and lenses, random superposition (each trap's phase drawn at random) and the random mask (each pixel assigned to one trap) are the references for the other methods. On the calibration task of Di Leonardo et al. [2] they show the published order: superposition is efficient only by accident and nearly non-uniform, random superposition gains efficiency at a spread, and the random mask has an efficiency of about the inverse of the number of traps. **Gerchberg–Saxton family (M04 to M06).** GS iterates between the SLM and the trap fields; the generalized adaptive additive form (GAA) adapts the update; weighted GS (GSW) rescales the target weights by the ratio of the mean to the achieved trap amplitude, $w_m \leftarrow w_m \langle r \rangle / r_m$, and reaches uniformity close to one [2]. **Direct search (M09)** minimizes the cost $C = -\langle I \rangle + f\sigma$ pixel by pixel over eight phase levels [5]. **Zernike correction (M14)** recovers injected aberrations by a sensorless modal search [10]. **Laguerre–Gaussian vortex traps (M15)** wind the ring phase by $2\pi l$. **Multi-plane GS (M19)** propagates plane by plane with an amplitude constraint per plane [11].

3.2 State-of-the-art tier

Phase-fixed weighted GS (M07) freezes the focal-plane phase after N iterations so that later iterations adjust only amplitudes [3]; the source reports 1500 foci at 91.2 percent modulation efficiency and camera

correction to 1.59 percent in three steps against 6.47 percent for conventional weighted GS. DinHot finds that the source default of $N = 12$ stalls on a crowded 50×30 array and uses $N = 60$ there (Table 3). **MRAF (M08)** frees a region outside the signal and trades efficiency for accuracy on shaped potentials [4]. **Conjugate-gradient minimization (M10)** uses powered intensity costs and an analytic gradient obtained from two FFTs [6]. **Wirtinger gradient descent with Adam (M11)** differentiates any forward model through the complex gradient [7]. **Closed-loop weighted GS (M12)** corrects trap amplitudes from a camera; DinHot runs it on an emulated camera whose errors are stated parameters, after Nogrette et al. [8]; the emulator reproduces the published feedback curves in the browser within the factor set in the design document (Table 3) and serves as a validation tool, not as a proposal. **Device-aware conjugate gradient (M13)** places the crosstalk kernel in the loop [9]. **Hungarian assignment and linear phase interpolation (M16)** sorts atoms with a cost $\sum d^{\alpha}$, $\alpha > 1$, which avoids crossings, and moves them along interpolated holograms [12, 13]. **An unsupervised convolutional network refined by GS (M17)** follows DeepCGH [14]. **Optimal-transport initialization of MRAF (M18)** seeds the phase from a transport map [15, 16].

Of the 24 engine methods, 24 ran in the bake of the cases: every one of them ran in at least one case. The two learned slots answer with the released graph (M17, on C14) or, with no model released, with their baseline (B3; Section 6).

Table 2: The method ladder of the design document: identifier, tier, method and acceptance criterion.

Id	Tier	Method	Acceptance criterion
M01	Classical	Superposition of prisms and lenses	on the 10x10 task with a flat beam, efficiency within 0.10 of 0.29 and uniformity below 0.05
M02	Classical	Random superposition	on the 10x10 task with a flat beam, efficiency above M01 and at least 0.65; uniformity below 0.35 (DinHot measured 0.25 at 256 squared against 0.01 at 768 squared in the source; the ordering, not the value, transfers)
M03	Classical	Random mask	efficiency about 1/M; at the source's 768 squared grid, uniformity within the spread of 0.58
M04	Classical	Gerchberg-Saxton, point-trap and FFT forms	the two forms agree to 10^{-9} rad; 30 iterations on the 10x10 task with a flat beam reach efficiency at least 0.9 and uniformity near 0.6
M05	Classical	Generalized adaptive additive	uniformity above GS at equal iterations on the 10x10 task
M06	Classical	Weighted GS	efficiency and uniformity at least 0.9 within a median of 15 iterations on the 10x10 task with a flat beam (the source reports about 10); the 3x3x3 lattice above 0.9 and 0.98
M07	State of the art	Phase-fixed weighted GS	fewer iterations than M06 to a relative standard deviation of 0.5 percent on a 50x30 array, with the phase fixed after 60 iterations in the acceptance run (the code default 'n_fix' is 12, as in Kim et al. 2019; at 12 it stalls at 2.9 percent on DinHot's grid); from the same start M07 matches slmsuite 0.5.0's WGS-Kim to 1e-4 percent (the first field is not reweighted, g_0 = 1)
M08	State of the art	MRAF	on disc, ring and bar targets, signal-region error at least five times below GS at the cost of efficiency
M09	Classical	Direct search	uniformity at least 0.98 on the 10x10 task; under 5 percent rms variation on a quasi-periodic set
M10	State of the art	Conjugate-gradient minimization	the analytic gradient matches finite differences; a square lattice below 1 percent error
M11	State of the art	Wirtinger gradient descent with Adam	lower trap-intensity error than M04 at equal iterations on the 10x10 task
M12	State of the art	Closed-loop WGS on an emulated camera	from a 19 percent rms open-loop spread, at most 2.1 percent in 20 iterations at gain 0.7; the acceptance runs on the emulated camera ('core/camera.py'), which reproduces the published feedback curves (Nogrette 2014: 19 percent to under 2.1; Kim 2019: 22 percent to under 1.8 in three phase-fixed steps)

Id	Tier	Method	Acceptance criterion
M13	State of the art	Device-aware CG with pixel crosstalk	on the crosstalk device, lower error than M10 without the model on a large disc; a staircase line search lets it run its budget on a quantizing device, where its scored spread does not improve systematically (a measured negative)
M14	Classical	Zernike correction and sensorless modal recovery	recovers an injected set of Noll modes 4 to 11 within 5 percent of their amplitude
M15	Classical	Laguerre-Gaussian vortex traps	the ring phase winds by 2 pi l with an error under 0.1 rad
M16	State of the art	Hungarian assignment and linear phase interpolation	no crossing moves for alpha 1.5 on a 7x7 to 3x3 sort; a moving trap above 70 percent of its mean intensity on the frames themselves (the minimum over the frames of the sequence over their mean; between frames, under the blend model of B2's transient, it is not: notebook 07-04)
M17	State of the art	Multi-plane CNN hologram with an optional volumetric GS refinement (DeepCGH-style)	on 162 held-out three-plane volumes, the hybrid (the network's phase, refined by M19 inside the same wall-clock budget only when that raises the volume accuracy) reaches a higher accuracy than M19 alone at that budget (measured median ratio 1.140); the refinement was kept on none of the volumes, so the measured method is the network alone
M18	State of the art	Optimal-transport initialized MRAF	lower error than MRAF from a random start on the shaped targets
M19	Classical	Multi-plane GS	on a three-plane target, every plane mean within 25 percent of the overall mean (volumetric 3D GS of Whyte and Courtial)
BS-1	Beyond	Device-aware weighted GS (engine id B1)	on the crosstalk device, relative standard deviation at least two times below M07, efficiency at most 10 percent lower; measured against the corrected M07 and against slmsuite's WGS-Kim alike: a median of 25.8 times (15.9 to 33.3 over five seeds) at 0.995 of the efficiency, and 2.0 times on the full C08 device (the bake, three seeds; 2.2 in notebook 03-04) with 4-bit phase and flicker
BS-2	Beyond	Flicker-bounded phase-continuous sequence (engine id B2)	minimum transient intensity of a moving trap at least 0.9 of its mean, against LPI on the same assignment: holds when the peak step per frame is at most 0.2 Fourier units, on most seeds (acceptance task: B2 0.916 to 0.926, LPI 0.551 to 0.709; long lattice moves: median 0.933 and 0.936, 6 of 8 seeds); on C10's long sort at 25 frames LPI leads in every seed, and B2 needs the end phases and 137 frames to reach 0.902 to 0.917 on 5 of 6 seeds of M16's plan (median 0.911; LPI 0.360 to 0.690), with one seed collapsing on its phase path
BS-3	Beyond	Learned warm start for phase-fixed WGS (refuted) (engine id B3)	measured negative: on held-out geometries the network start needs a median 8 iterations to 1 percent against 14 from a random start (1.75 times, below the 2 times claimed), and a control with the same early phase fix and no network gains as much; the model is not released and the claim BS-3 is reported as refuted
BS-4	Beyond	Aberration-robust hologram (engine id B4)	over a 200-sample ensemble of residual aberrations, the 95th percentile of the relative standard deviation at least 1.25 times below M07, efficiency at most 10 percent lower (measured: a median of 1.435 times, 1.42 to 1.57 over three seeds, at 0.982 of the efficiency); robustness to pointing errors is measured and reported, not claimed (limited by the spot width: 1.04 times)
BS-5	Beyond	Efficiency-uniformity Pareto front (engine id B5)	hypervolume at least that of the M04, M05, M09 and M06 points on the 10x10 task

Table 3: Measured acceptance of the methods on DinHot's own grids (unit tests; seeds as stated). Intervals are [min, max].

Id	Measured	Seeds
M01	efficiency 0.281, uniformity 0.0003 (deterministic)	1
M02	efficiency 0.750 [0.738, 0.752], uniformity 0.252 [0.106, 0.272]	6
M03	256 squared: efficiency 0.010, uniformity 0.057 [0.031, 0.072]; 768 squared: efficiency 0.008, uniformity 0.566 [0.543, 0.590]	5 and 2
M04	flat beam: efficiency 0.939 [0.933, 0.939], uniformity 0.604 [0.557, 0.626]; Gaussian beam: efficiency 0.815 against its ceiling 0.869	3
M05	uniformity 0.688 [0.638, 0.709] against 0.615 [0.550, 0.637] for GS at equal iterations; efficiency 0.811 against 0.816	5
M06	flat beam: efficiency and uniformity both at 0.9 after 12, 15 and 12 iterations (median uniformity 0.857 at 10); Gaussian beam: 12, 15 and 12; 3 x 3 x 3 lattice (0.9 and 0.98): 35, 20 and 27; budgets 15, 20 and 100	3
M07	50 x 30 array to 0.5 percent in 64 and 64 iterations against 136 and 106 for M06 (phase fixed after 60; efficiency 0.83 and 0.83); with the source value of 12 it stalls at 3.58 percent	2
M08	signal-region error 12.4 (disc), 7.8 (ring) and 10.4 (bar) times below GS at the worse of two starts; efficiency 0.181 to 0.184 against 0.963 to 0.992	2 starts
M09	uniformity 0.9994 and 0.9992 (efficiency 0.66 and 0.65); 119 quasi-periodic traps at 0.023 percent rms	2 and 1
M10	analytic gradient within 2.5e-06 of finite differences; square lattice error 0.55 and 0.51 percent	2
M11	relative standard deviation 2.53, 2.46 and 2.26 percent against 17.78, 17.98 and 18.82 for GS at 30 iterations	3
M12	emulated camera loop: open-loop 20.9 and 20.4 percent to 0.84 and 1.07 percent in 20 iterations; 24.3 and 23.6 percent to 1.05 and 0.74 percent in three phase-fixed steps	2
M13	error 0.241 (efficiency 0.535) against 1.382 (0.507) for M10 without the device model, large disc on the crosstalk device	1
M14	worst per-mode relative error 0.002 [0.001, 0.002] over Noll modes 4 to 11	3
M15	ring phase winds by 2 pi l (winding error below the 1e-9 rad of the arithmetic) with a maximum error of 0.033 [0.028, 0.060] rad (isolated vortices of charge 1, 2, 3 and -2)	2 per charge
M16	0 path crossings in 60 loadings; moving traps keep 0.767 [0.747, 0.792] of their mean intensity (worst moving trap of each of 5 loadings)	60 loadings
M17	three-plane network refined by M19: median accuracy ratio 1.140 (quartiles 1.102 and 1.160) on 162 held-out volumes; taken from the committed metrics.json, not remeasured here	162 volumes
M18	signal-region error 0.0204 (disc), 0.0179 (ring) and 0.0221 (line) against 0.250, 0.188 and 0.223 for MRAF from a random start (the better of two); baked size: 0.0259 (disc) and 0.0171 (ring) against 0.266 and 0.228	2 and 1
M19	plane means over overall mean 0.938, 0.987 and 1.075 and 0.926, 1.034 and 1.039 on three planes	2
BS-4	95th percentile of the relative standard deviation 3.947 against 6.192 for M07 (1.57 times) over a 200-sample aberration ensemble; efficiency 0.7047 against 0.7181; pointing errors 1.04 times	1
BS-5	hypervolume 0.8190 against 0.8018 for the union of the M04, M05, M06 and M09 points; 7 front points dominate M04 and 8 dominate M05; front built in 48 seconds	1

4 The cases

A case is an optical setup, a device, a target (a list of traps or a shaped potential), an aberration, a method set, a seed and, for dynamic cases, a particle and a rearrangement. There are 14 cases in four categories (Table 4): arrays (a 3×3 grid, a 10×10 grid with camera feedback, a 32×32 array of 1024 traps, the calibration task and a disordered array of 819 traps on integer FFT bins at least five bins apart), shaped and three-dimensional targets (a disc potential with the crosstalk model, a three-plane array with vortices, a vortex ring with a line, and a three-plane volume at the geometry of the released learned model), device and optics (high numerical aperture, a device-limited array, an aberrated array with and without correction) and dynamics and controls (an atom-rearrangement sequence and a trap outside the field, which is refused and never computed). Every case states numeric expected ranges for its metrics, measured on three seeds, and the bake fails outside them. All cases are synthetic: the parameters are stated and no measured sample is used. The baked cases span 9 to 1024 traps.

Table 4: The fourteen cases: category, optical setup, target and the methods that ran in the bake. C11 is refused by the input contract and never computed.

Case	Category	Setup	Target	Methods	Reference
C01	Arrays	850 nm, NA 0.5	9 traps	14	M06
C02	Arrays	850 nm, NA 0.5	100 traps	14	M07
C03	Arrays	850 nm, NA 0.5	1024 traps	7	M07
C04	Shaped and 3D	850 nm, NA 0.5	disc potential, 8-bit	6	M13
C05	Shaped and 3D	1064 nm, NA 1.2	51 traps	10	M09
C06	Shaped and 3D	1064 nm, NA 1.2	25 traps	8	M07
C07	Device and optics	1064 nm, NA 1.2	25 traps	13	M07
C08	Device and optics	850 nm, NA 0.5	64 traps, 4-bit, fill 0.85	7	B1
C09	Device and optics	850 nm, NA 0.5	16 traps, aberrated	6	M14
C10	Dynamics and controls	850 nm, NA 0.5	100 traps	4	M16
C11	Dynamics and controls		one trap at 200 μm, outside the field	0	refused
C12	Arrays	850 nm, NA 0.5	100 traps	7	M06
C13	Arrays	850 nm, NA 0.5	819 traps	7	M07
C14	Shaped and 3D	850 nm, NA 0.5	48 traps	8	M17

Table 5: Calibration case C12 (10×10 traps at eight Fourier units): efficiency e and uniformity u of the analytic and Gerchberg–Saxton baselines as baked, against the values published by Di Leonardo et al. for the same task.

Method	e (baked)	u (baked)	σ (%)	e (published)	u (published)
M01	0.328	0.000	268.79	0.29	0.01
M02	0.749	0.199	37.40	0.69	0.01
M03	0.010	0.077	54.25	0.01	0.58
M04	0.817	0.506	19.97	0.94	0.60
M05	0.818	0.738	11.81	0.93	0.79
M06	0.805	0.970	1.02	0.93	0.99
M07	0.797	1.000	0.01	n/a	n/a

5 Protocols and results

5.1 Baked cases

Each case is computed offline with every method valid for it (the seed is stated in each manifest), checked against its expected ranges, and committed as a manifest and a trace of at most 256 KiB. Table 5 compares the calibration case with the published rows of Di Leonardo et al.: the ordering of the six baselines and the weighted-GS uniformity are reproduced; the efficiencies of the GS family are lower than published because the engine's Gaussian input beam caps the efficiency, whereas the source uses a flat beam (the flat-beam test of M04 gives 0.939 efficiency and 0.604 uniformity; Table 3). The DinHot uniformity of random superposition and of the random mask differ from the published values on the engine's smaller grid, as the acceptance test of those methods states: the order, not the value, transfers.

Table 6 summarizes the baked cases. Where both ran, phase-fixed weighted GS (M07) leaves a median relative standard deviation of 0.01 percent against 18.7 percent for plain GS (M04) over 9 cases, at efficiencies between 0.453 and 0.817. On the device-limited case C08 phase-fixed weighted GS (M07), which ignores the device, keeps a spread of 13.4 percent, and the device-aware weighted GS (B1) lowers it to 7.1 percent, a factor of 1.9, on that small grid (the larger task of Section 7 gives a larger factor; both are reported in the companion manuscript). On the aberrated case C09, correction of the Zernike modes (M14) restores efficiency and uniformity that the uncorrected methods lose.

5.2 Feedback and particles

The emulated camera loop reduces the spread as in the published curves. In the baked traces the relative standard deviation falls from 9.1 to 0.02 percent in 5 steps on C01, from 20.1 to 1.34 percent in 20 steps on the 10×10 grid C02, from 22.3 to 1.44 percent in 3 steps on the large array C03, and from 22.0 to 0.44 percent in 5 steps on the disordered array C13 (Fig. 4). The published closed-loop result of Nogrette et al. [8] is a trap-depth spread of 19 percent rms reduced to 1.4 percent in 20 iterations; the unit test of M12 reproduces it within the factor set in the design document (Table 3). The particle simulations act on the computed traps: C01 carries a bead in the central trap (stiffness 4.01 pN/μm and a corner frequency of 70.0 Hz in the trace) and C03 an atom (a corner frequency of 107200 Hz and a radial trap frequency of 109092

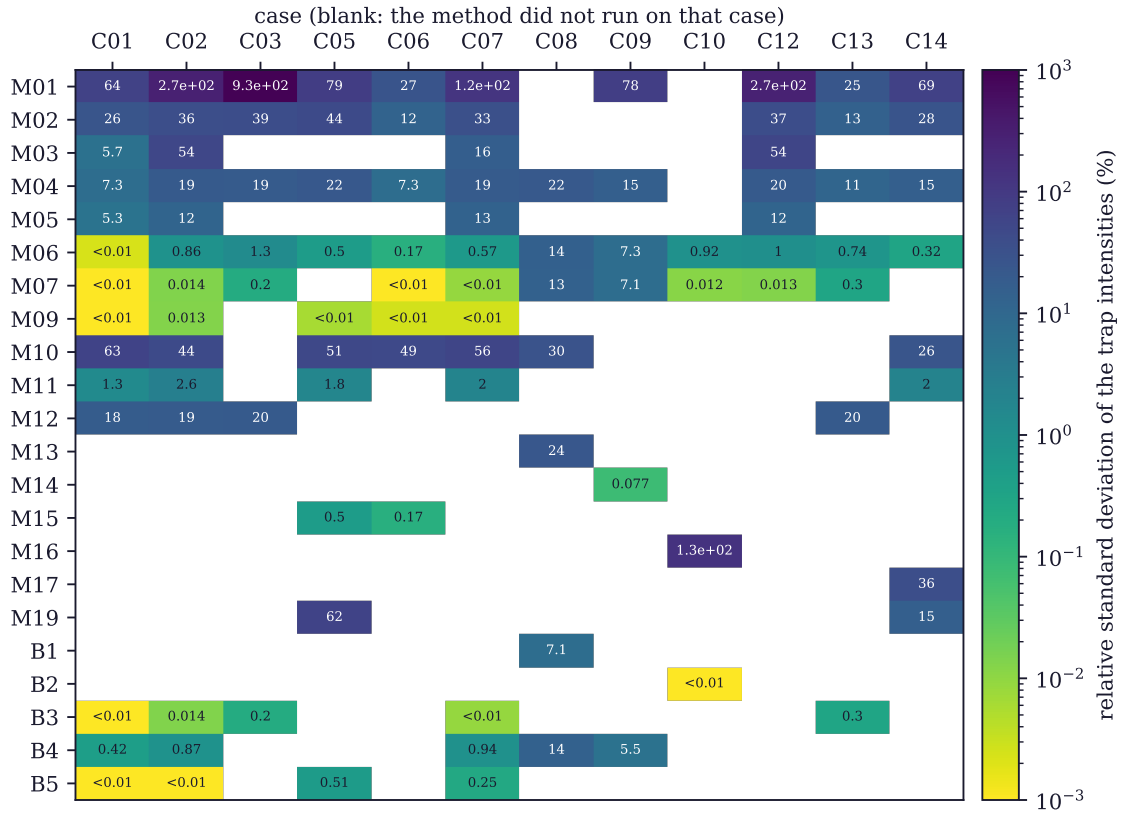


Figure 2: Relative standard deviation of the trap intensities, every method on every point-trap case of the bake (the shaped-potential case C04 is scored on its signal region and is omitted). Each cell is one method run at the case budget. Plain Gerchberg–Saxton (M04) leaves the largest spreads; the weighted and phase-fixed forms (M06, M07), the direct search (M09), the Pareto front (B5) and the device-aware methods remove most of it where they apply.

Hz in the trace); the trap depth follows the intensity [19], and the bead's spectrum is a Lorentzian whose corner frequency is the stiffness over $2\pi\gamma$ [20], which the particle oracle checks.

5.3 The benchmark protocol

The benchmark compares the tiers on one protocol. Its geometries are (7) a 10×10 square, a ring of 36 traps, 100 random traps, a quasi-periodic set of 119, a $5 \times 5 \times 3$ lattice, a large array (32×32 and the disordered 819 set) and shaped fields; the grids are 128, 256 pixels, the iteration budgets are 10, 30, 100 and the time budgets are 0.05, 1 s (the last is the converged row). A group runs up to 20 seeds, at least two, until a bound on its wall clock, and every cell reports the median and the inter-quartile range of efficiency, uniformity, relative standard deviation, time and iterations, on the ideal device and on the C08 device (every hologram scored against its device). The Pareto fronts are stored with the cells.

The committed results file holds 1326 rows of 21 of the 24 engine methods; 61 method–geometry cells are recorded as skipped with the reason the file states (a method that does not accept the target, such as a two-dimensional method on a three-dimensional lattice, or one that refuses the geometry). A cell runs between 2 and 20 seeds, the bound the protocol puts on the wall clock of a group. The 3 methods without a row (M15, M16, M17) refuse every geometry of the protocol by their specification (a vortex charge, a rearrangement block, the geometry the learned model was trained for); their tests and their cases measure them. Table 7 names, per geometry and device, the method of each tier with the lowest median spread at 100 iterations among those that reach at least half of the cell's best efficiency, and the best method at a time budget of 1 s; Table 8 gives the medians of every method over the 6 point-trap geometries, and Fig. 5 every cell.

On the ideal device the best method of a geometry is classical on 1, state of the art on 0 and beyond on 4 of the 7 geometries, and on 2 the best methods of two or more tiers tie below the resolution of the file (0.001 percent); on the C08 device the counts are 2, 3 and 1 of 6 (the multi-plane geometry has no device rows),

Table 6: Baked results per case at the case budget: the method with the lowest relative standard deviation among those with at least half the best efficiency, and the phase-fixed WGS (M07) and plain GS (M04) rows where they ran. e efficiency, σ relative standard deviation of the trap intensities (%).

Case	Best	e	σ	M07 σ	M07 e	M04 σ
C01	B5	0.817	<0.01	<0.01	0.817	7.3
C02	B5	0.798	<0.01	0.01	0.798	19.5
C03	M07	0.794	0.20	0.20	0.794	18.7
C04	shaped potential	n/a	n/a	n/a	n/a	n/a
C05	M09	0.699	0.01	n/a	n/a	22.4
C06	M07	0.704	<0.01	<0.01	0.704	7.3
C07	M09	0.638	<0.01	0.01	0.791	18.7
C08	B1	0.652	7.05	13.42	0.675	21.6
C09	M14	0.805	0.08	7.09	0.453	15.0
C10	B2	0.798	<0.01	0.01	0.797	n/a
C12	M07	0.797	0.01	0.01	0.797	20.0
C13	M07	0.723	0.30	0.30	0.723	11.2
C14	M06	0.785	0.32	n/a	n/a	15.3

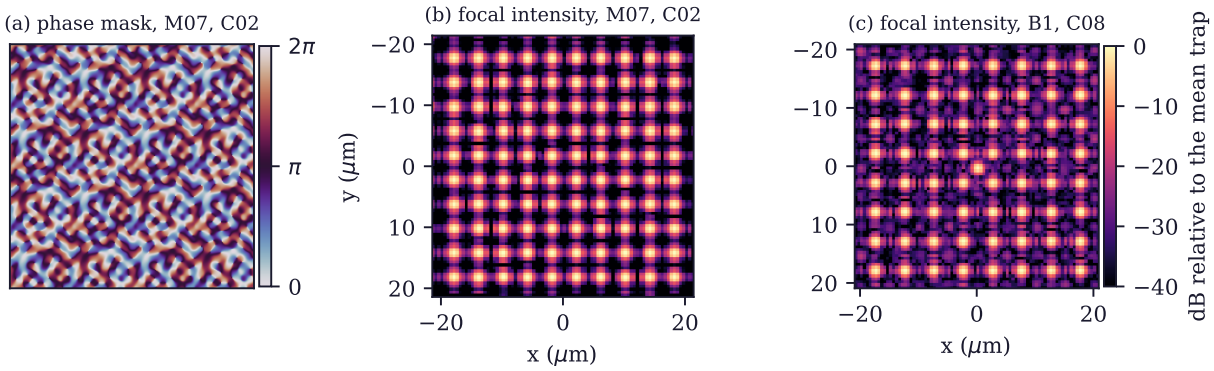


Figure 3: (a) Phase mask of the reference method on C02. (b) Its focal intensity, in dB relative to the mean trap, on the ideal device: a regular lattice. (c) The focal intensity of the device-aware weighted GS on C08, scored on the device with 4-bit phase, crosstalk and a fill factor of 0.85: the traps are visible and not equal.

with 0 ties. B2 enters these cells as its static frame, a phase-locked weighted GS, and not as a sequence, which is the quantity of its claim (Section 7). Over the point-trap geometries the lowest median spread on the ideal device is that of BS-2 (<0.001 percent at a median efficiency of 0.756), and on the C08 device that of M11 (13 percent at 0.511). The medians of the 17 methods that keep at least half of the best efficiency span <0.001 to 31 percent on the ideal device and 13 to 49 percent (15 methods) on the C08 device, where every method runs with its default parameters and only the device-aware ones model the device. The times are wall-clock seconds of the machine that ran the benchmark, under shared load.

6 The learned models

Two networks were trained offline with PyTorch, exported to ONNX with a list of operators the browser runtime supports, and evaluated on held-out sets written before the final training with other seeds. A model is released only if it meets its criterion; otherwise the baseline answers.

cgh-net (M17). A U-Net maps three plane amplitudes, at −8, 0 and +8 SLM-limited Rayleigh ranges, to one phase for the 128×128 SLM, in the setting of DeepCGH [14]. It is trained without labels through the differentiable FFT model, with the loss $1 - \mathrm{AC}$ over the planes, $\mathrm{AC} = 2\sum TI/(\sum T^2 + \sum I^2)$ on unit-sum plane intensities, for 16000 steps. The exported graph is 4.67 MB. The baseline is volumetric three-dimensional GS (M19) on the same traps with a wall-clock budget of 7.35 s, the median time of M19 at 30 iterations. On 162 held-out volumes the network answers in about 35 ms with a median accuracy ratio of 1.140 (quartiles 1.102 and 1.160), above the baseline on 100 percent of the volumes; the median accuracy is 0.687 against 0.600. By kind the ratio is 1.162 (lattices), 1.147 (random), 1.118 (mixed), 1.100 (rings), 1.065 (the $5 \times 5 \times 3$ benchmark lattice) and 1.188 (the three-plane layout). The ONNX graph agrees with PyTorch on the reconstructed plane intensities to 8.4×10^{-6} (relative to the maximum). The accuracy measure

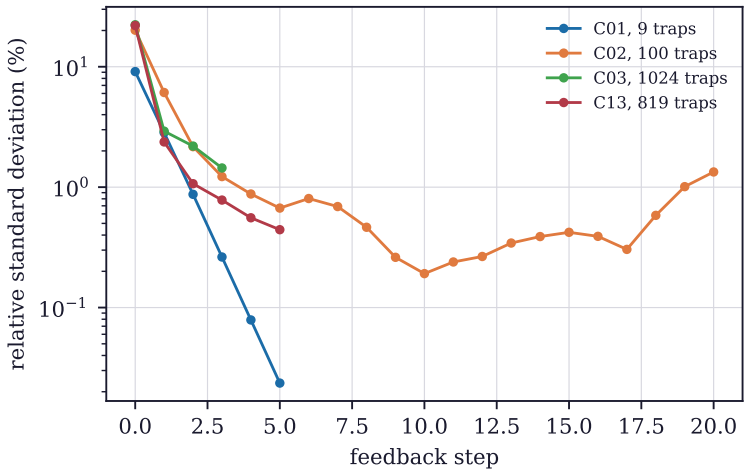


Figure 4: Emulated camera feedback: relative standard deviation of the trap intensities against the feedback step, on four cases (baked traces).

Table 7: Benchmark, per geometry and device: the method of each tier with the lowest median relative standard deviation σ (%) at 100 iterations among those that reach at least half of the best median efficiency of the cell, and the best method of all at a time budget of 1 s. Default parameters, medians over the seeds of each cell.

Geometry	Device	Classical	State of the art	Beyond	Best at 1 s
G1 Square 10 x 10	ideal	M09 0.066	M11 1.2	BS-1 <0.001	BS-5 <0.001
G1 Square 10 x 10	C08	M09 24	M11 13	BS-4 24	BS-5 25
G2 Ring of 36 traps	ideal	M09 0.015	M07 <0.001	BS-1 <0.001	BS-1 <0.001
G2 Ring of 36 traps	C08	M01 6.6	M13 8.9	BS-1 7.4	M01 6.6
G3 Random 100 traps	ideal	M09 0.052	M07 <0.001	BS-1 <0.001	BS-1 <0.001
G3 Random 100 traps	C08	M14 20	M11 15	BS-1 4.5	BS-4 20
G4 Quasi-periodic 119	ideal	M09 0.071	M07 0.012	BS-1 <0.001	BS-5 <0.001
G4 Quasi-periodic 119	C08	M05 84	M11 21	BS-4 84	BS-1 28
G5 3D lattice 5 x 5 x 3	ideal	M09 0.05	M11 2.2	BS-2 <0.001	BS-2 <0.001
G6 Large: 32 x 32 and disordered 819	ideal	M09 0.15	M07 0.75	BS-1 <0.001	BS-4 1
G6 Large: 32 x 32 and disordered 819	C08	M09 19	M11 13	BS-2 19	BS-4 19
G7 Shaped fields (ring, disc, line, LG ring)	ideal	M04 35	M18 5.6	n/a	M18 36
G7 Shaped fields (ring, disc, line, LG ring)	C08	M04 40	M11 30	n/a	M18 39

scores full plane images, ghosts and blur included, which is what the network optimises; M19 optimises trap uniformity, so the ratio is a statement about this measure. The two-dimensional counterpart, a single-plane network followed by GS refinement, ties: on 219 held-out targets the ratio at equal wall-clock is 0.9994, below the criterion, and that model is not released.

warmstart-model (BS-3). A permutation-equivariant set network predicts initial trap phases for phase-fixed WGS. On 66 held-out geometries it needs a median of 8 iterations to reach 1 percent against 14 from a random start (a factor of 1.75, below the factor of two set as the criterion), but only with the early phase freeze (after 3 iterations, against 12 for the random arm). A control with no network, a random start and the same early freeze, needs a median of 9 iterations on the control subset, and 40 percent of its geometries never reach 1 percent (47 percent with the network); the final efficiency differs by 4.2 percent between the arms. The gain comes from the freeze, not from the network, so the proposal is refuted and the model is not released (Fig. 6).

7 Proposals beyond the published state of the art

Each proposal is DinHot's own, grounded in a stated gap, measured against the strongest baseline and reported whether or not it succeeds.

BS-1, device-aware weighted GS. The weights of weighted GS are kept and the phase update is replaced by conjugate-gradient steps through the crosstalk kernel and the quantizer. The gap is that the crosstalk model has only been used with conjugate gradient on shaped potentials [9], or inside a weighted GS that replaces amplitudes only [17]. On the crosstalk device, the relative standard deviation is lower than that of

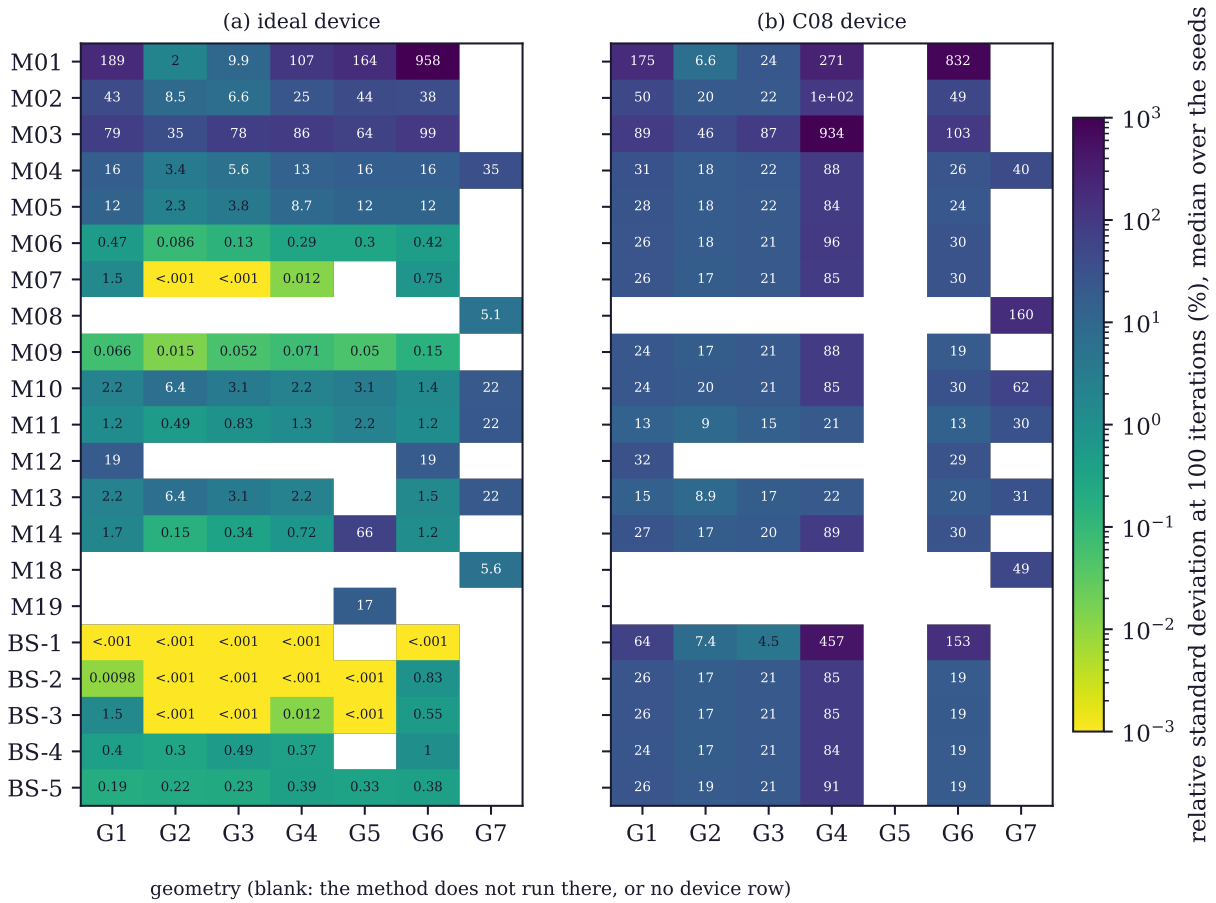


Figure 5: Benchmark: median relative standard deviation at 100 iterations with the default parameters, every method on every geometry, on the ideal device (a) and on the C08 device (b). Blank: the method does not accept the geometry or refuses it, or the geometry has no device row.

phase-fixed WGS by the factor reported in the companion manuscript; on the small C08 grid the baked factor is 1.9 at an efficiency ratio of 0.966. **BS-2, flicker-bounded phase-continuous sequences.** Each intermediate hologram of a moving-trap sequence is optimized with the trap phases locked to a smooth trajectory; the gap is that linear phase interpolation does not optimize its intermediate frames [13]. The moving traps keep a higher minimum intensity than under linear phase interpolation, as quantified in the companion manuscript. **BS-3, learned warm start:** refuted (Section 6). **BS-4, aberration-robust hologram:** the expected cost over residual aberrations is minimized; over a 200-sample ensemble of residual Noll modes 4 to 11 the 95th percentile of the relative standard deviation is 1.57 times below that of M07 on the seed of its test, with an efficiency 1.9 percent lower (Table 3); robustness to pointing errors does not follow (1.04 times), because the spot width bounds it, and is reported as a negative result. The gap is that corrections are global or per site, never robust [18]. **BS-5, efficiency–uniformity Pareto front** by continuation of a weighted cost: the front's hypervolume (0.8190) exceeds that of the points of the GS family and direct search (0.8018), and 7 front points dominate M04 and 8 dominate M05 (Table 3), where published methods report single trade-off points [2].

8 The live engine and the product

The site is static; every computation runs in the browser or was baked offline and is replayed. The live engine is a TypeScript port of the forward models, the device model and the live methods, running in a Web Worker so the interface does not block, with the FFT in Float64 (fft.js). An optional WebGPU backend runs the FFT-form iterations on the GPU and falls back to the CPU path when the browser offers none; an agreement check inside every GPU run compares the trap fields with the CPU result. At the live grid (256^2 padded to 512^2) a weighted GS iteration measured about 24 ms, so a 30-iteration run fits the lane budget of 1500 ms. The Python engine writes 46 reference documents and 92 parity tests compare the TypeScript

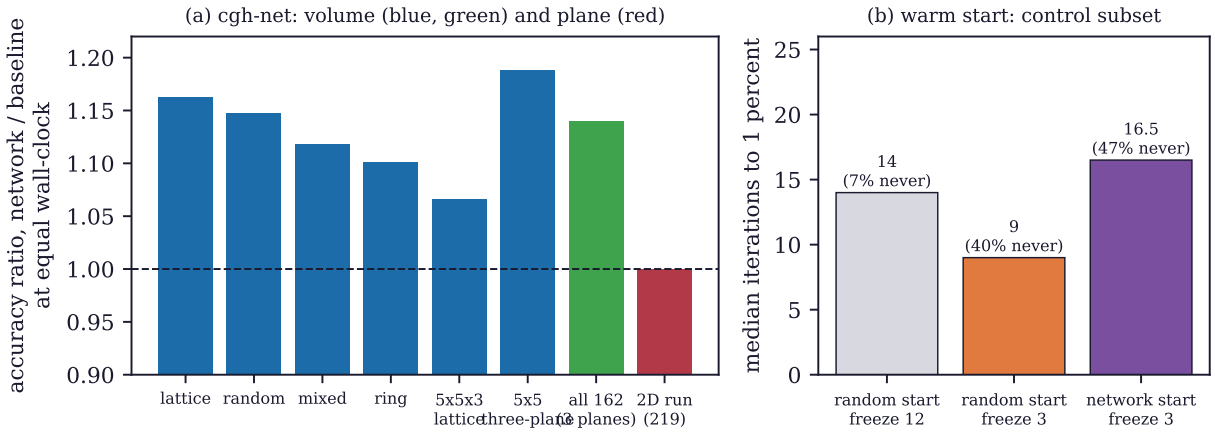


Figure 6: (a) The cgh-net accuracy ratio over its baseline at equal wall-clock, by target kind on the held-out volumes (G5: the $5 \times 5 \times 3$ lattice; 3-plane: the 5×5 three-plane layout), for the volume network (blue, green) and the two-dimensional single-plane network (red); the dashed line is the baseline. (b) The warm-start control: median iterations to 1 percent for a random start with the original and the early freeze and for the network start with the early freeze.

engine with them, each quantity within the tolerance its document states (at most 10^{-9} absolute). On an integrated Intel gen-12lp adapter, 30 iterations of M07 took 0.31 to 0.32 s with the GPU backend against 1.5 to 1.6 s on the CPU path in the same browser. The methods that are baked and replayed are labelled as such in the interface. Of the 11 data contracts of the build, 7 are fully available, and the build declares the state of the others: `params-record` is partial (1 of 14 records rejected by contract 1); `benchmark-results` is partial (61 of 168 benchmark cells were not run); `warmstart-model` is training (No released model; the slot's baseline answers (GS for cgh-net, random trap phases for warmstart-model)); `api` is missing (The API lane is dormant; the site reads the committed documents). The refused record is the case outside the field, and the benchmark counts its skipped cells as not run; the dormant API serves GET requests and answers 503 with the declared state.

9 Limits

DinHot computes and evaluates holograms in simulation: it does not drive an SLM or a camera, and its camera-in-the-loop method runs on an emulated camera whose errors are stated parameters. Device and set-up parameters come from published experiments, so absolute numbers belong to the stated model and not to a measured instrument. The engine is scalar (Fraunhofer and angular spectrum); the high-numerical-aperture case carries a validity flag and the vectorial lane is not built. The fill-factor, 4-bit and flicker effects of the device have no published source in this build and are computed only. The crosstalk kernel's frequency unit is carried over from the engine and is not verified against the source. The transient-intensity model of BS-2 is the engine's own (Section 7 and the companion manuscript) and the linear-phase-interpolation baseline is DinHot's implementation of a published description, not the authors' code. The unit measurements of Table 3 use between one and six seeds; the benchmark repeats the methods on its geometries with 2 to 20 seeds per cell. The cgh-net advantage is on a plane-image accuracy measure over generated volumes and the benchmark's volumetric layouts, not on a hardware experiment. The learned warm start is refuted and the two-dimensional network ties with GS. The executive figures of the introduction are taken from the cited publications and not re-measured.

Data and code availability

The source code, the baked artifacts (manifests, traces, benchmark results, model cards, the ONNX graph and its held-out metrics) and the scripts that regenerate every table and figure of this report from them are in the repository `https://github.com/fsantibanezleal/CEFOP_DinHot` (Apache-2.0), under `manuscripts/holographic-trap-studio/`; the studio runs at `https://dinhot.fasl-work.com`. Unit measurements quoted in Table 3 are produced by the tests named in Table 2.

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Table 8: Benchmark, per method: medians over the point-trap geometries the method ran on (of 6) at 100 iterations with the default parameters, on the ideal device and on the C08 device. σ relative standard deviation (%), e efficiency, time wall-clock seconds under shared load.

Id	Tier	Geometries	σ ideal	e ideal	time (s)	σ C08	e C08
M01	Classical	6	136	0.399	0.0082	175	0.267
M02	Classical	6	31	0.722	0.01	49	0.478
M03	Classical	6	78	0.016	0.098	89	0.013
M04	Classical	6	14	0.799	1.8	26	0.523
M05	Classical	6	10	0.798	1.6	24	0.514
M06	Classical	6	0.3	0.790	1.6	26	0.518
M07	State of the art	5	0.012	0.751	1.2	26	0.503
M08	State of the art	0	n/a	n/a	n/a	n/a	n/a
M09	Classical	6	0.059	0.642	2.6e+02	21	0.431
M10	State of the art	6	2.6	0.751	1.7	24	0.486
M11	State of the art	6	1.2	0.776	1.1	13	0.511
M12	State of the art	2	19	0.822	8.8	30	0.592
M13	State of the art	5	2.2	0.737	2	17	0.485
M14	Classical	6	0.94	0.742	0.89	27	0.511
M18	State of the art	0	n/a	n/a	n/a	n/a	n/a
M19	Classical	1	17	0.892	21	n/a	n/a
BS-1	Beyond	5	<0.001	0.751	53	64	0.296
BS-2	Beyond	6	<0.001	0.756	3.5	21	0.503
BS-3	Beyond	6	0.006	0.754	1.6	21	0.503
BS-4	Beyond	5	0.4	0.744	34	21	0.498
BS-5	Beyond	6	0.28	0.774	25	21	0.512